



SE-356 Game Application Development

SPACE ARENA

Submitted by

Muhammad Umer Luqman

FALL-2022-BSSE-026-A

Submitted to

Ma'am Kiran Amjed

**Department of Software Engineering,
Lahore Garrison University, Lahore.**

1. Game Overview

Space Arena is a 2D top-down space shooter built in Unity. The player controls a spaceship at the center of the screen and must survive waves of enemies approaching from multiple directions. The game features a 3-level progression system where enemy speed, spawn rate, and attack directions increase with each level. The player earns points by destroying enemies with lasers and loses lives when enemies reach the player. The game ends when all three lives are lost.

Property	Details
Language	C# (MonoBehaviour)
Perspective	2D Orthographic — Top-Down
Total Scenes	2 — Main_Menu (index 0), Game (index 1)
Total Scripts	11 C# Scripts
Levels	3 progressive levels in one Game scene
Player Lives	3 — Game Over when all lost
Score Per Kill	10 points per enemy
Audio	laser_shot.wav (SFX), tintin.mp3 (BGM)

2. Scene Structure

Scene	Scripts Active	What Happens
Main_Menu (0)	MainMenu.cs	Shows game logo and Play button. Clicking Play calls SceneManager.LoadScene(1) to start the game.
Game (1)	All 10 remaining scripts	Full gameplay: player movement, shooting, 3-level progression, enemy spawning, scoring, HUD, and Game Over state.

3. Level Progression System

Space Arena uses a score-based level progression system managed by LevelManager.cs. The game starts paused with a Level 1 popup. As the player scores points, the game automatically transitions to harder levels with a popup announcement and a brief pause between levels. Each level increases enemy speed and spawn frequency.

Level	Unlock Score	Enemy Speed	Spawn Rate (Top)	New Feature
Level 1	0 (Start)	3 units/sec	Every 2.5 sec	Top enemies only. Popup: 'Top enemies incoming!'

Level	Unlock Score	Enemy Speed	Spawn Rate (Top)	New Feature
Level 2	Score $\geq$ 50	5 units/sec	Every 1.2 sec	Faster enemies. Popup: 'Enemy spawn speed increased!'
Level 3	Score $\geq$ 100	6 units/sec	Every 0.9 sec	Bottom enemies added (every 1.5 sec). Popup: 'Bottom enemies coming!'

3.1 Level Transition Workflow

1. Player score reaches threshold (50 or 100)
2. LevelManager.OnScoreChanged() detects the threshold
3. SpawnManager.StopSpawning() halts all coroutines
4. SpawnManager.ClearAllEnemies() destroys all active enemies
5. GameManager.SetPaused(true) freezes player input
6. UiManager.ShowLevelPopup() displays level title + message
7. Player presses SPACE or ENTER to continue
8. UiManager.HideLevelPopup() removes overlay
9. GameManager.SetPaused(false) resumes player input
10. SpawnManager.BeginLevel(n) starts new spawn settings

4. Scripts & Components

Space Arena uses 11 C# scripts. Each script has a single clear responsibility. The table below shows every script, the GameObject it is attached to, and what it does.

Script	Attached To	Responsibility
MainMenu.cs	Menu Button	Loads Game scene when Play is clicked via SceneManager.LoadScene(1).
Player.cs	Player GameObject	Reads H/V input for movement, fires lasers on Spacebar, tracks lives and score, notifies UI and level manager.
Laser.cs	Laser Prefab	Moves laser upward at 8 units/sec, destroys self when off-screen top ($y \geq 7$).
Enemy.cs	Enemy Prefab (Top)	Moves downward. Destroyed at $y \leq -7$ or on collision with PLAYER/LASER tags.
BottomEnemy.cs	BottomEnemy Prefab	Moves upward. Destroyed at $y \geq 7$ or on collision. Active only in Level 3.
LeftEnemy.cs	LeftEnemy Prefab	Moves rightward at 4 units/sec. Destroyed at $x \geq 9$ or on collision.
RightEnemy.cs	RightEnemy Prefab	Moves leftward at 4 units/sec. Destroyed at $x \leq -9$ or on collision.

Script	Attached To	Responsibility
SpawnManager.cs	SpawnEnemies GameObject	Manages enemy wave spawning per level. Runs coroutines for top and bottom enemies. Clears enemies on level transition.
LevelManager.cs	Game_Manager GameObject	Tracks current level. Listens to score changes. Triggers level transitions and popups at score 50 and 100.
GameManager.cs	Game_Manager GameObject	Tracks game-over and pause states. Handles R key restart. Exposes IsPaused and IsGameOver flags to other scripts.
UiManager.cs	Canvas GameObject	Updates score text, lives sprite, Game Over text, restart text. Builds and shows/hides the level popup panel at runtime.

5. Player System

5.1 Movement & Controls

Action	Input	Behaviour
Move Left / Right	H Axis (A/D or Arrow Keys)	Translates player on X axis at 6 units/sec via Time.deltaTime
Move Up / Down	V Axis (W/S or Arrow Keys)	Translates player on Y axis at 6 units/sec via Time.deltaTime
Fire Laser	Spacebar	Instantiates laser prefab at player position, plays laser_shot.wav
Confirm Level Popup	Space / Enter	Dismisses level popup and resumes gameplay
Restart Game	R (Game Over only)	SceneManager.LoadScene(1) — only works when _isGameOver = true

5.2 Player Boundaries

Axis	Clamp Range	Behaviour
X (Horizontal)	-8 to +8	Player position is hard-clamped each frame. Cannot move past left or right edges.
Y (Vertical)	-4 to +4	Player position is hard-clamped each frame. Cannot move past top or bottom edges.

5.3 Lives & Damage System

State	What Happens
Lives = 3 (Start)	UiManager shows 3-lives sprite. Full health.
Enemy hits Player	player.damage() → _lives-- → UiManager.updateLives() swaps sprite
Lives = 2	2-lives sprite shown on HUD (_spriteImg[2])
Lives = 1	1-life sprite shown on HUD (_spriteImg[1])
Lives = 0	0-lives sprite shown, Game Over sequence triggered, player destroyed, spawning stopped

6. Enemy System

6.1 Enemy Types

Enemy Type	Direction	Speed	Destroyed At	Active From
Enemy (Top)	Vector3.down	LevelManager speed	y <= -7	Level 1
BottomEnemy	Vector3.up	LevelManager speed	y >= +7	Level 3 only
LeftEnemy	Vector3.right	4 units/sec (fixed)	x >= +9	Not yet in spawn system
RightEnemy	Vector3.left	4 units/sec (fixed)	x <= -9	Not yet in spawn system

6.2 Enemy Collision Logic (All Types)

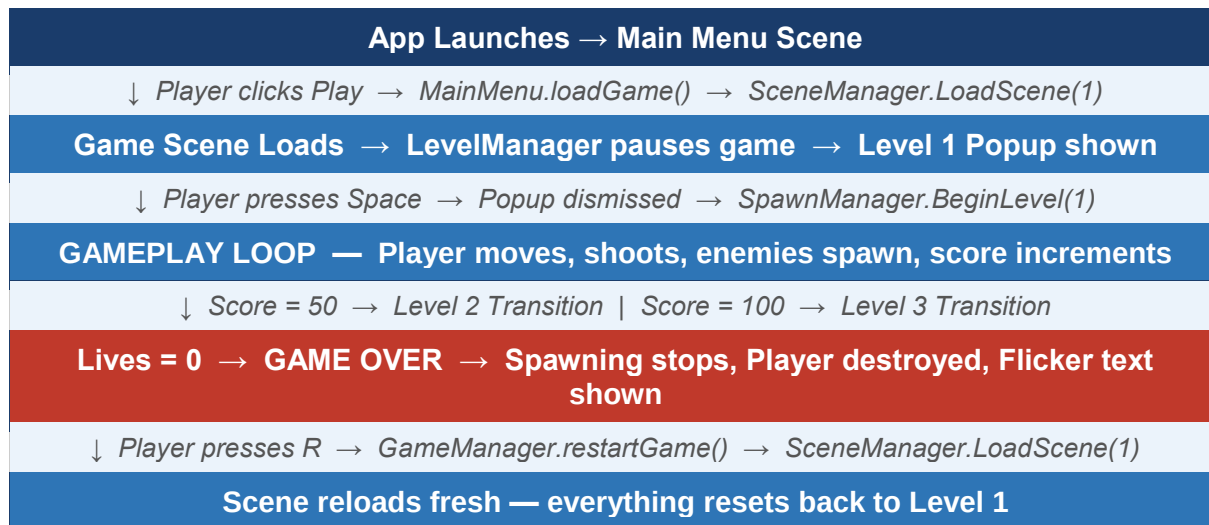
Collision Tag Detected	Action Taken
PLAYER	Calls _player.damage() (if player not null), then Destroy(this enemy). Player loses 1 life.
LASER	Destroy(this enemy) + Destroy(laser) + _player.addScore(10). Score increases by 10.
Off-Screen Boundary	Destroy(this enemy). No damage, no score — enemy simply missed the player.

6.3 Spawn Manager — Per-Level Settings

Level	Top Spawn Interval	Bottom Spawn Interval	Active Coroutines
Level 1	2.5 seconds	Not active	spawningFromTheTop() only
Level 2	1.2 seconds	Not active	spawningFromTheTop() only (faster)
Level 3	0.9 seconds	1.5 seconds	spawningFromTheTop() + spawningFromTheBottom()

7. Game Workflow

7.1 Complete Game Flow



7.2 Script Communication Map

From Script	To Script	Method / Action
Any Enemy Script	Player	damage() and addScore(10)
Player	UiManager	updateLives() and updateScore()
Player	LevelManager	OnScoreChanged(score)
Player	SpawnManager	onPlayerDeath()
LevelManager	SpawnManager	StopSpawning(), ClearAllEnemies(), BeginLevel(n)
LevelManager	UiManager	ShowLevelPopup(), HideLevelPopup()
LevelManager	GameManager	SetPaused(true/false)
UiManager	GameManager	gameOver()
GameManager	SceneManager (Unity)	LoadScene(1) on R key press
MainMenu	SceneManager (Unity)	LoadScene(1) on Play button click

8. UI System

UI Element	Type	What It Does
scoreText	UI Text	Displays 'Score : X'. Updated every time addScore() is called.
_livesImg	UI Image	Shows lives sprite. Swaps from index 3→2→1→0 as lives decrease.
gameOverText	UI Text	Hidden at start. Shown on Game Over. Flickers on/off every 0.5 sec via Coroutine.
restartText	UI Text	Hidden at start. Shows 'Press R to Restart' on Game Over.
LevelPopupPanel	GameObject (built at runtime)	Semi-transparent black overlay with level title, description, and 'Press SPACE to continue' prompt.
_levelTitleText	UI Text (in popup)	Shows level name e.g. 'Level 1', 'Level 2', 'Level 3'.
_levelMessageText	UI Text (in popup)	Shows level description e.g. 'Top enemies incoming!'
_levelContinueText	UI Text (in popup)	Shows 'Press SPACE to continue' in yellow colour.

9. Unity Components Used

Component	Used On	Role in Space Arena
MonoBehaviour	All 11 scripts	Base class providing Start(), Update(), and Coroutine support
Transform	All GameObjects	Position, rotation, scale. Translate() used for all movement.
Collider2D (IsTrigger)	Player, Enemies, Laser	Trigger overlap detection. Fires OnTriggerEnter2D on contact.
Rigidbody2D	Player, Enemies	Required for 2D trigger detection to work in Unity.
SpriteRenderer	All visual objects	Renders the 2D sprite image on screen.
AudioSource	Player GameObject	Plays laser_shot.wav SFX on Spacebar press.
AudioClip	Player (SerializeField)	The .wav file assigned in Inspector for laser fire sound.
Canvas + UI Text	Canvas GameObject	Displays score, lives, Game Over text, restart prompt.
UI Image	Canvas GameObject	Displays lives sprite (swapped based on remaining lives count).
SceneManager	MainMenu + GameManager	Loads scenes by build index. Used for Play and Restart.
Coroutine / IEnumerator	SpawnManager, UiManager, LevelManager	Runs timed background tasks: enemy spawning, flicker effect, level popup wait loop.
Time.deltaTime	Player, all enemy scripts	Multiplied into speed for frame-rate independent movement.
Random.Range()	SpawnManager	Generates random X spawn positions for enemies.
GameObject.Find()	Most scripts	Locates GameObjects in scene by name to get script references.
Instantiate() / Destroy()	Player, SpawnManager, All enemies	Creates and removes prefab instances at runtime.

10. Audio & Asset Reference

10.1 Audio

File	Format	Trigger
laser_shot.wav	WAV	Plays via AudioSource.Play() every time player presses Spacebar
tintin.mp3	MP3	Background music — plays on scene load via AudioSource Play On Awake

10.2 Key Sprites

Sprite	Usage
Player.png	Player spaceship sprite rendered by SpriteRenderer
Enemy.png	Top enemy sprite
BottomEnemy.png	Bottom enemy sprite (different visual from top)
laser.png	Laser projectile sprite
UI/Lives/Three.png	_spriteImg[3] — shown at 3 lives (full health)
UI/Lives/Two.png	_spriteImg[2] — shown after first hit
UI/Lives/One.png	_spriteImg[1] — shown after second hit
UI/Lives/no_lives.png	_spriteImg[0] — shown at 0 lives, triggers Game Over

11. Game Parameters Summary

Parameter	Value
Player Move Speed	6 units/sec (H and V axis × Time.deltaTime)
Player Boundary X	-8 to +8 (hard clamp)
Player Boundary Y	-4 to +4 (hard clamp)
Laser Speed	8 units/sec (Vector3.up)
Laser Destroy Boundary	Y >= 7 (off-screen top)
Enemy Speed — Level 1	3 units/sec (LevelManager.EnemyMoveSpeed)
Enemy Speed — Level 2	5 units/sec
Enemy Speed — Level 3	6 units/sec
Top Spawn Rate — Level 1	Every 2.5 seconds
Top Spawn Rate — Level 2	Every 1.2 seconds
Top Spawn Rate — Level 3	Every 0.9 seconds
Bottom Spawn Rate — Level 3	Every 1.5 seconds
Level 2 Unlock Score	50 points
Level 3 Unlock Score	100 points
Score Per Kill	10 points
Player Lives	3
Game Over Flicker Rate	0.5 seconds on / 0.5 seconds off
Restart Key	R (only active when _isGameOver = true)

— End of Document —